Acoustic Emission and Passive Acoustic Regression to the ISO 281 Viscosity Ratio κ for Online, Non-Invasive Lubrication Condition Monitoring of Ball Bearings: An Exploratory Study

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Abstract

Inadequate lubrication is responsible for an estimated 30–80% of rolling element bearing failures, yet reliable online, non-invasive lubrication condition monitoring (LCM) remains an unsolved engineering problem. Passive sensor-based approaches are most practical for industrial retrofit because they require no bearing modification, but the signal processing challenge of separating lubrication-relevant information from confounded operating-condition effects has not been systematically characterised for simultaneous acoustic emission (AE) and passive acoustic sensing. This paper presents an exploratory study in which AE and passive acoustic signals are simultaneously acquired from a ball bearing across two temperature set-points (40 °C and 75 °C) and five rotational speed levels (500–2500 rpm), covering the boundary, mixed, and full-film lubrication regimes. The ISO 281 viscosity ratio $\kappa \in [0.16, 1.55]$ is derived analytically from bearing geometry, lubricant properties, and operating conditions as a continuous, physics-grounded regression target. Twenty-six time-domain and frequency-domain features are extracted per sweep and evaluated by both raw and partial Spearman rank correlation, the latter controlling for rotational speed and temperature. In the raw analysis, Hjorth mobility is the top AE predictor ($|\rho| = 0.69$) and broadband spectral standard deviation leads for the acoustic channel ($|\rho| = 0.58$); however, the partial analysis reveals that these correlations are substantially driven by operating-condition effects. After speed and temperature control, frequency-kurtosis and normalised bandwidth emerge as the most robust AE indicators of κ ($\rho_{\text{partial}} \approx +0.23$), while acoustic-channel complexity and spectral shape features carry the strongest independent information ($\rho_{\text{partial}} \approx +0.39$). Four regression model families are benchmarked across three sensor configurations; gradient-boosted regression on AE features alone achieves the best hold-out performance ($R^2 = 0.64$, RMSE $= 0.19$). Fusing AE and acoustic features does not improve upon single-modality AE ($R^2 = 0.60$ combined vs. 0.64 AE alone), a finding attributed to the limited independent information carried by the heterodyned acoustic channel. These results identify spectral shape and frequency-distribution features as physically motivated, operating-condition-robust κ indicators and establish a first benchmark for simultaneous dual-channel passive sensing regressed to κ. The dataset is made publicly available to support standardised comparison.

Contents

1 Introduction

Rolling element bearings are among the most prevalent mechanical components in industrial machinery, and their reliable operation depends critically on adequate lubrication. Lubrication failure—whether caused by starvation, viscosity degradation, or contamination—increases metal-to-metal asperity contact, accelerates surface fatigue and wear, raises operating temperature, and ultimately leads to premature bearing failure. Improper lubrication is estimated to account for between 30% and 80% of all rolling element bearing failures [1–3].

Despite this, lubrication condition monitoring (LCM) has received substantially less research attention than vibration-based fault detection and remaining useful life estimation, which are now mature technologies deployed widely in industry. The principal reason is that lubrication-related signal changes are subtler and more entangled with operating conditions than the impulse signatures of surface spalling or raceway cracks. Accurate, online LCM would enable condition-based relubrication: applying lubricant at the right time in the right quantity rather than following time-based schedules that systematically produce both under- and over-lubrication [4].

Passive, non-invasive sensing methods—acoustic emission (AE), vibration, and passive ultrasound—are the most practically attractive for industrial LCM because they require no bearing modification and can be retrofitted to bearings in service. Among these, AE is sensitive to the broadband stress waves emitted by asperity contact events and microcrack propagation across a wide frequency range (typically 50 kHz–4 MHz), while passive ultrasound at 38–40 kHz is already industrially deployed for threshold-based lubrication assessment [5]. Despite both being in practical use, they have not been studied simultaneously in a unified regression framework with a physics-based lubrication reference variable.

A key advance in the field has been the use of the ISO 281 viscosity ratio κ as a continuous regression target [1, 6]. Since κ encodes the ratio of actual lubricant viscosity to the minimum required for full elastohydrodynamic lubrication (EHL) at the current operating conditions, a model trained to predict κ implicitly learns the operating-condition dependence and provides a physically interpretable, standardised output. Jakobsen et al. [1, 6] demonstrate κ regression from vibration and passive ultrasound features using LASSO and neural networks, establishing the methodological foundation that this paper builds upon.

The complementarity of AE and passive acoustic sensing for LCM has been argued on physical grounds in several reviews [7, 8], but has not been quantified experimentally in a unified framework. Whether jointly exploiting both sensor modalities improves κ prediction over either alone is an open question with direct practical implications: additional sensors impose cost and complexity, and are only justified if they provide non-redundant information.

Contributions

This paper makes the following contributions:

1. **Novel dual-channel experiment.** The first study to simultaneously acquire AE and passive acoustic signals from a ball bearing and regress both, individually and jointly, to κ using a common dataset and analytical ground truth.

2. **Partial Spearman feature analysis.** A systematic evaluation of 26 features by raw and partial Spearman rank correlation (controlling for speed and temperature) distinguishes amplitude-dominated features from operating-condition-robust shape features, and reveals divergent behaviour between the two sensor modalities.

3. **Quantified fusion test.** The first quantitative test of AE–acoustic complementarity in a unified experimental framework, including a negative fusion result that directly addresses an open question in the LCM literature.

4. **Open dataset.** A publicly available simultaneous AE + passive acoustic bearing dataset with analytically derived κ reference values, addressing the absence of standardised benchmark data identified as a primary gap in the field [7].

The remainder of the paper is organised as follows. Section 2 reviews the theoretical background of κ, the Hamrock–Dowson model, and the physical origins of AE and acoustic signals. Section 3 describes the experimental setup and data collection. Section 4 presents the signal processing and feature extraction pipeline. Section 5 presents the feature analysis results including partial Spearman correlations. Section 6 presents the regression modelling results. Section 7 discusses the findings. Section 8 concludes.

2 Background

2.1 Lubrication regimes and the viscosity ratio κ

The lubrication regime of a rolling element bearing is determined by the ratio of the elastohydrodynamic (EHL) film thickness h_m to the composite surface roughness σ of the contact surfaces, quantified by the specific film thickness $\Lambda = h_m/\sigma$. When $\Lambda \geq 3$ the contact is in the full-film EHL regime; when $0.5 \leq \Lambda < 3$ mixed lubrication prevails with partial asperity contact; when $\Lambda < 0.5$ boundary lubrication dominates with predominantly metal-to-metal contact.

The ISO 281 standard defines the viscosity ratio

$$\kappa = \frac{\nu_0}{\nu_1(n,\, d_{pw})}, \tag{1}$$

where ν_0 [$\mathrm{mm}^2.\mathrm{s}^{-1}$] is the actual kinematic viscosity of the lubricant at operating temperature, and ν_1 is the minimum kinematic viscosity required for full EHL at the bearing's rotational speed n [rpm] and mean pitch diameter d_{pw} [mm] [9]. The reference viscosity is given by

$$\nu_1 = \begin{cases} 45\,000\; n^{-0.83}\; d_{pw}^{-0.5} & n < 1000\,\mathrm{rpm}, \\ 4\,500\; n^{-0.5}\; d_{pw}^{-0.5} & n \geq 1000\,\mathrm{rpm}. \end{cases} \tag{2}$$

The operating viscosity ν_0 is computed from the Walther viscosity–temperature equation fitted to the lubricant's 40 °C and 100 °C reference values. A value of $\kappa \geq 1$ indicates adequate film formation; $\kappa < 0.5$ corresponds to severe boundary lubrication.

Since κ encodes both n and T through ν_1 and ν_0, it cannot be varied independently of operating conditions without changing the lubricant itself. Critically, using κ as a regression target—rather than a discrete fault label—means that a trained model implicitly compensates for operating-condition effects: the model learns the full mapping from raw features to the physics-grounded EHL adequacy index.

2.2 Physical origins of AE and passive acoustic signals

I don't know if this section should be moved to the signal processing section. The origin of the signals is what motivates the choice of fea-

Acoustic emission (AE) refers to transient elastic stress waves generated by sudden energy releases in a material: asperity contact and micro-fracture are the dominant sources in rolling bearings. Under boundary and mixed lubrication, intermittent metal-to-metal asperity contacts generate broadband stress waves that propagate through the bearing structure and are detected by piezoelectric AE sensors coupled to the outer housing. The emission rate, amplitude, and frequency content of these events are all sensitive to the contact severity and hence to the lubrication regime [10, 11]. Full-film EHL suppresses asperity contact, reducing AE activity; the transition through the mixed regime is associated with intermittent, structurally distinct emission events. AE sensors typically operate in the 50 kHz–1 MHz range and require sampling rates exceeding 500 kHz to capture the full frequency content.

Passive ultrasound sensing at 38–40 kHz provides a complementary, lower-frequency view of bearing acoustics. Commercial instruments (UE Systems, Sonotec) compare the broadband ultrasonic level against a reference baseline, triggering relubrication when the level rises by a specified number of decibels above the reference [5]. This threshold approach is the only passive non-invasive LCM method in routine industrial deployment but is confounded by speed and load changes. In the present experiment the passive acoustic channel uses a heterodyned probe, which frequency-shifts the signal into the audible range and effectively limits the usable bandwidth to approximately 10 kHz; this constitutes a different signal from wideband AE and from classical ultrasonic reflectometry.

The two sensor modalities are therefore sensitive to partially overlapping and partially distinct physical phenomena. AE captures high-frequency transient stress waves from individual asperity contacts, while the passive acoustic channel reflects lower-frequency bulk acoustic behaviour and machinery noise components. Whether this partial independence translates into complementary κ information is the central empirical question of this study.

2.3 Hjorth parameters and spectral shape features

Hjorth parameters are time-domain features derived from the signal's variance and its successive derivatives [12]. Mobility M is defined as

$$M = \sqrt{\frac{\mathrm{Var}(\dot{x})}{\mathrm{Var}(x)}}, \tag{3}$$

where $\dot{x}$ is the time derivative of signal x. Mobility quantifies the mean frequency content of the signal: high mobility indicates a higher proportion of rapid transitions, consistent with impulsive asperity contact events. Complexity $C = M(\dot{x})/M(x)$ quantifies the deviation from a pure sinusoid.

Spectral shape features—frequency kurtosis, normalised bandwidth, spectral flatness—describe the distribution of signal energy across the frequency axis independently of absolute energy level. Unlike amplitude features (RMS, peak, variance), spectral shape features are invariant to overall amplitude scaling and are therefore less directly confounded by rotational speed.

3 Experimental Setup

3.1 Test rig and bearing

The test rig is a custom-built bearing test stand at CeramicSpeed (Holstebro, Denmark). A single deep-groove ball bearing (model SYJ25, bore 25 mm, pitch diameter d_{pw} = 38.0 mm, 9 chrome steel rolling elements, dynamic load rating 14000 N) is driven by a variable-frequency electric motor controlled through a variable-frequency drive (VFD). The bearing is loaded radially by a static weight and lubricated by a defined quantity of Keratech 22 oil (kinematic viscosity $\nu_{40} = 22$ cSt, $\nu_{100} = 4.1$ cSt, NLGI grade 1 equivalent, ISO VG 22). Temperature is monitored by a PT100 sensor on the bearing housing and controlled passively through the test duration and ambient conditions.

Note on VFD noise: Harmonic artefacts from the VFD switching frequency are present at approximately 5 kHz and its multiples in the low-frequency range of both sensor signals. A replacement VFD with filter is planned for subsequent experiments; in the present analysis, low-frequency spectral features below 10 kHz should be interpreted with this artefact in mind.

3.2 Sensor configuration and data acquisition

Two sensors are acquired simultaneously:

AE sensor. A wideband piezoelectric AE sensor is bonded to the bearing housing. Sampling rate: 1.6 MHz (nominally; effective bandwidth verified to 800 kHz by the DAQ). Signal recorded to a digital oscilloscope with voltage range adapted to the signal level (500 mV–1 V per division depending on RPM; see Table 1).

Passive acoustic sensor. A heterodyned passive ultrasound probe (commercial instrument) is clamped to the bearing housing. The heterodyning process frequency-shifts the raw acoustic signal into the audible range (< 10 kHz); the recorded signal therefore reflects low-frequency acoustic emissions rather than true wideband ultrasound. Voltage range: 50 mV per division. Sampling rate: 1.6 MHz (recorded at the same rate as the AE channel; effective content below 10 kHz).

Data are acquired in individual time-series sweeps triggered at each operating condition. Each sweep comprises a fixed number of samples from both channels recorded simultaneously. The two sensors share identical acquisition timing, ensuring that feature comparisons and correlation analyses are conducted on co-temporal signal segments.

3.3 Operating conditions and κ derivation

Table 1: Experimental operating conditions and resulting mean κ values. Counts are per-sensor (AE or acoustic); each sweep row contributes one AE and one acoustic record. The mixed-regime gap at $\kappa \approx 0.5$–0.7 for intermediate RPM/temperature combinations reflects the discrete temperature set-points used.

RPM	**Temp (°C)**	**Mean κ**	$n_{\mathbf{sweeps}}$	**Regime**
500	40	0.47	218	Boundary
500	75	0.17	198	Boundary
1000	40	0.75	26	Mixed
1000	50	0.68	96	Mixed
1000	75	0.30	102	Boundary
1500	40	1.09	220	Full film
1500	50	0.87	74	Mixed
1500	75	0.39	104	Boundary
2000	40	1.10	18	Full film
2000	50	1.01	70	Full film
2000	75	0.44	94	Boundary
2500	40	1.25	32	Full film
2500	50	1.12	68	Full film
2500	75	0.49	100	Boundary

The experimental design covers two nominal temperature set-points (40 °C and 75 °C) and five rotational speed levels (500–2500 rpm), yielding $\kappa \in [0.16, 1.55]$ with mean $\bar{\kappa} =$

0.61 and standard deviation $\sigma_\kappa = 0.33$. Regime distribution across the 799 AE sweeps retained after cleaning: 852 boundary sweeps ($\kappa < 0.5$, 53%), 390 mixed sweeps ($0.5 \leq \kappa < 1.0$, 24%), and 358 full-film sweeps ($\kappa \geq 1.0$, 23%). Data were collected across two experimental sessions (December 2025 and February 2026) stored in six HDF5 files totalling 1600 sweep–sensor pairs before quality filtering.

The κ value for each sweep is computed according to Equations (1)–(2) using the per-sweep measured RPM and temperature, with ν_0 evaluated via the Walther viscosity–temperature equation. No separate reference measurement of film thickness is available; κ is therefore a model-derived quantity subject to the assumptions of the Hamrock–Dowson framework and Walther's equation.

Note on operating condition coverage. The dataset has a gap in the mixed lubrication regime at intermediate κ values (≈ 0.5–0.7) caused by the discrete temperature set-points: at 40 °C and 75 °C most RPM levels fall in either the boundary or full-film regime. Model performance in the mixed regime is consequently evaluated on relatively few points, which is a limitation discussed in Section 7.

3.4 Dataset availability

The complete dataset (HDF5 signal files, extracted features in Parquet format, and the analysis pipeline scripts) will be made publicly available on Zenodo upon acceptance. The Python analysis pipeline is configured via a single YAML file and is fully reproducible from the raw HDF5 files.

4 Signal Processing Pipeline

In this section the signal processing pipeline applied to each sweep–sensor pair, comprising signal cleaning, and feature extraction.

- Goal of the pipeline
- Elements included in the pipeline
- Practical implementation (Python version, libraries used, etc.)

4.1 Signal cleaning

4.2 Filter bands

4.3 Feature extraction

Twenty-six features are computed from the cleaned voltage signal for each sweep–sensor pair, comprising 12 time-domain and 14 frequency-domain statistics, as listed in Table 2.

Table 2: Features extracted from each sweep–sensor pair. Notation: N signal length; $\bar{x}$ sample mean; σ population standard deviation; $S_k = |X_k|$ one-sided FFT magnitude at bin k; f_k corresponding frequency; K bin count; $\bar{S} = K^{-1}\sum_k S_k$; $\sigma_S = \mathrm{std}(\{S_k\})$; $\Sigma S = \sum_k S_k$; $f_c = (\Sigma S)^{-1}\sum_k f_k S_k$; $\sigma_f = (K^{-1}\sum_k (f_k - f_c)^2)^{1/2}$.

Name	**Formula**	**Description**
Time-domain		
Peak	$(\max x - \min x)/2$	Half the peak-to-peak range.
RMS	$\sqrt{N^{-1}\sum_n x_n^2}$	Root mean square amplitude.
Std	$\sqrt{N^{-1}\sum_n (x_n - \bar{x})^2}$	Population standard deviation.
Variance	$N^{-1}\sum_n (x_n - \bar{x})^2$	Population variance (square of Std).
Skewness	$N^{-1}\sum_n [(x_n - \bar{x})/\sigma]^3$	Third standardised central moment; amplitude asymmetry.
Kurtosis	$N^{-1}\sum_n [(x_n - \bar{x})/\sigma]^4$	Fourth standardised central moment; tail heaviness.
Crest factor	$x_{\mathrm{peak}}/x_{\mathrm{RMS}}$	Peak-to-RMS ratio; sensitive to impulsive events.
Shape factor	$x_{\mathrm{RMS}} \,/\, (N^{-1}\sum_n |x_n|)$	Ratio of RMS to mean absolute value.
Impulse factor	$x_{\mathrm{peak}} \,/\, (N^{-1}\sum_n |x_n|)$	Ratio of peak to mean absolute value.
Margin factor	$x_{\mathrm{peak}} \,/\, (N^{-1}\sum_n \sqrt{|x_n|})^2$	Peak divided by squared mean square-root amplitude.
Hjorth mobility	$[\mathrm{Var}(x')/\,\mathrm{Var}(x)]^{1/2}$	Square root of first-derivative to signal variance ratio; proportional to mean frequency.
Hjorth complexity	$M(x')/M(x)$	Ratio of first-derivative mobility M to signal mobility; bandwidth proxy.
Frequency-domain		
Dominant frequency	$f_{\arg\max_k S_k^2}$	Frequency at which spectral power is maximum.
Spectral mean	$\bar{S} = K^{-1}\sum_k S_k$	Arithmetic mean of the FFT magnitude spectrum.
Spectral std	$\sigma_S = \mathrm{std}(\{S_k\})$	Population standard deviation of the FFT magnitude spectrum.
Spectral skewness	$\sum_k (f_k - f_c)^3 S_k \,/\, (\sigma_S^3\, \Sigma S)$	Frequency asymmetry of the magnitude-weighted spectrum, normalised by magnitude std.
Spectral kurtosis	$\sum_k (f_k - f_c)^4 S_k \,/\, (\sigma_S^4\, \Sigma S)$	Frequency tail heaviness of the magnitude-weighted spectrum, normalised by magnitude std.

(Table 2 continued)

Name	Formula	Description
Spectral flatness	$\exp(K'^{-1}\sum_{k:S_k>0}\ln S_k) \,/\, \bar{S}$	Geometric-to-arithmetic mean ratio of the spectrum; 1 for white noise, 0 for a pure tone.
Centre frequency	$f_c = \sum_k f_k S_k \,/\, \Sigma S$	Magnitude-weighted mean frequency (spectral centroid).
RMS frequency	$(\sum_k f_k^2 S_k \,/\, \Sigma S)^{1/2}$	Magnitude-weighted root mean square frequency.
Freq.-weighted std	$\sigma_f = (K^{-1}\sum_k (f_k - f_c)^2)^{1/2}$	RMS deviation of bin frequencies from centre frequency (unweighted by magnitude).
Peak frequency	$(\sum_k f_k^4 S_k \,/\, \sum_k f_k^2 S_k)^{1/2}$	RMS frequency computed with $f_k^2 S_k$ weighting.
Norm. freq. std	$\sigma_f \,/\, f_c$	Frequency standard deviation normalised by centre frequency.
Frequency skewness	$\sum_k (f_k - f_c)^3 S_k \,/\, (K\sigma_f^3)$	Third spectral moment of frequency deviation from centroid, weighted by magnitude.
Frequency kurtosis	$\sum_k (f_k - f_c)^4 S_k \,/\, (K\sigma_f^4)$	Fourth spectral moment of frequency deviation from centroid, weighted by magnitude.
Norm. bandwidth	$\sum_k \lvert f_k - f_c\rvert^{1/2} S_k \,/\, (K\sigma_f^{1/2})$	Half-order moment of spectral spread, normalised by $\sigma_f^{1/2}$.

All frequency-domain features are computed from the one-sided FFT magnitude spectrum of the cleaned signal.

4.4 Feature selection

Two criteria are applied to reduce the 26-feature space before modelling:

1. **Spearman rank correlation with κ**: features are ranked by $|\rho(\text{feature}, \kappa)|$. The ranking is used for analysis only; all 26 features proceed to the redundancy filter.
2. **Variance inflation factor (VIF) and pairwise correlation filter**: features with VIF > 10 or more than three pairwise correlation partners ($|r_{ij}| > 0.8$) are flagged as redundant and removed by a greedy procedure that retains the highest-$|\rho|$ representative from each correlated group.

After selection, 10 AE features are retained: *skewness, crest factor, kurtosis, shape factor, mobility, complexity, dominant frequency, centre frequency, RMS frequency, peak* *frequency.* Eleven acoustic-channel features are retained: *skewness, crest factor, kurtosis, shape factor, mobility, complexity, dominant frequency, spectral standard deviation, centre frequency, spectral flatness, peak frequency.* All retained features are normalised to zero mean and unit variance before modelling.

5 Feature Analysis

5.1 Raw Spearman correlation with κ

Table 3 lists the top ten features for each sensor ranked by $|\rho(\text{feature}, \kappa)|$ across all 799 sweeps per sensor.

Table 3: Top 10 features by raw Spearman correlation $|\rho|$ with κ for AE and acoustic channels. Features above the horizontal rule are in the retained set used for modelling.

AE		**Acoustic**	
Feature	$\lvert\rho\rvert$	Feature	$\lvert\rho\rvert$
Hjorth mobility	0.691	Spectral std	0.580
Spectral flatness	0.616	RMS	0.573
Variance	0.561	Hjorth mobility	0.561
Std	0.561	Std	0.560
RMS	0.557	Variance	0.560
Peak	0.554	Peak	0.511
Spectral std	0.552	Freq. kurtosis	0.496
Hjorth complexity	0.493	Norm. bandwidth	0.496
Freq. kurtosis	0.449	Freq. skewness	0.496
Norm. bandwidth	0.449	Spectral mean	0.496

In the raw analysis, amplitude features (RMS, variance, peak) and Hjorth mobility dominate both sensors, with $|\rho| > 0.55$ across the board. Figure 1 illustrates the Hjorth mobility vs. κ relationship for both sensors. Whilst the negative trend is visually apparent, the scatter is large and the clusters correspond closely to operating-condition groups, suggesting that operating speed is a dominant driver.

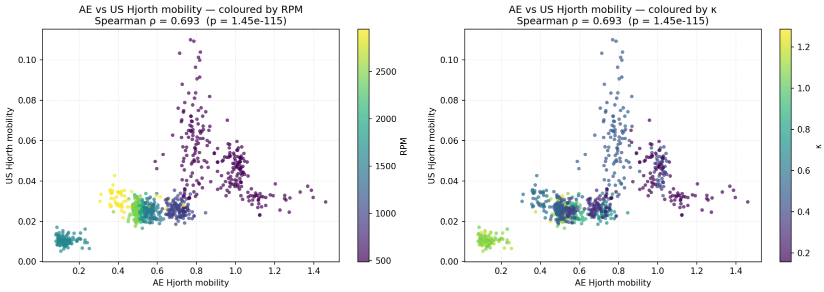

Figure 1: Hjorth mobility vs. κ for AE (left) and passive acoustic (right) sensors coloured by rotational speed. The visible clustering by RPM indicates that a substantial fraction of the apparent κ–mobility relationship is mediated by operating speed.

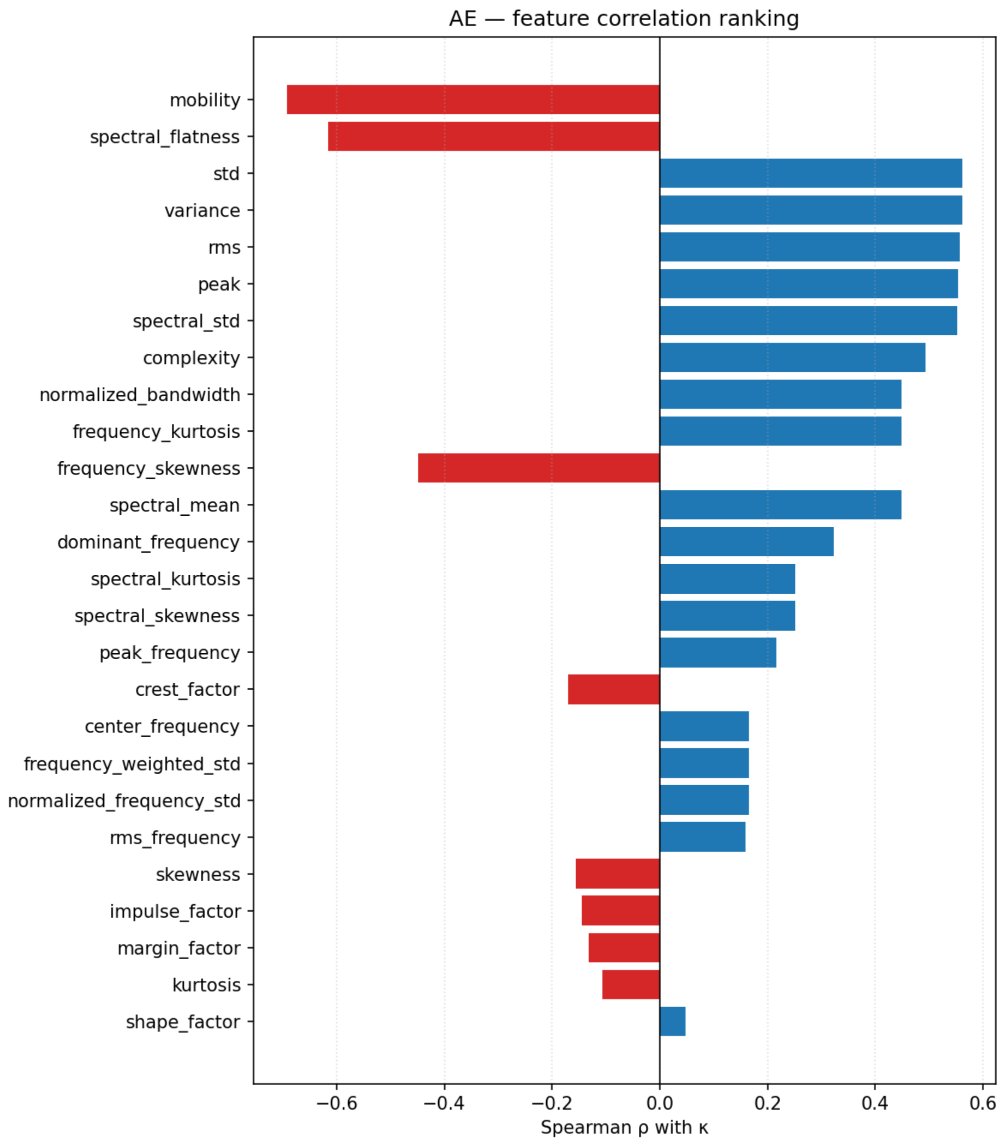

Figure 2: Spearman $|\rho|$ between each AE feature and κ, sorted by magnitude. Retained features (after VIF and redundancy filtering) are highlighted.

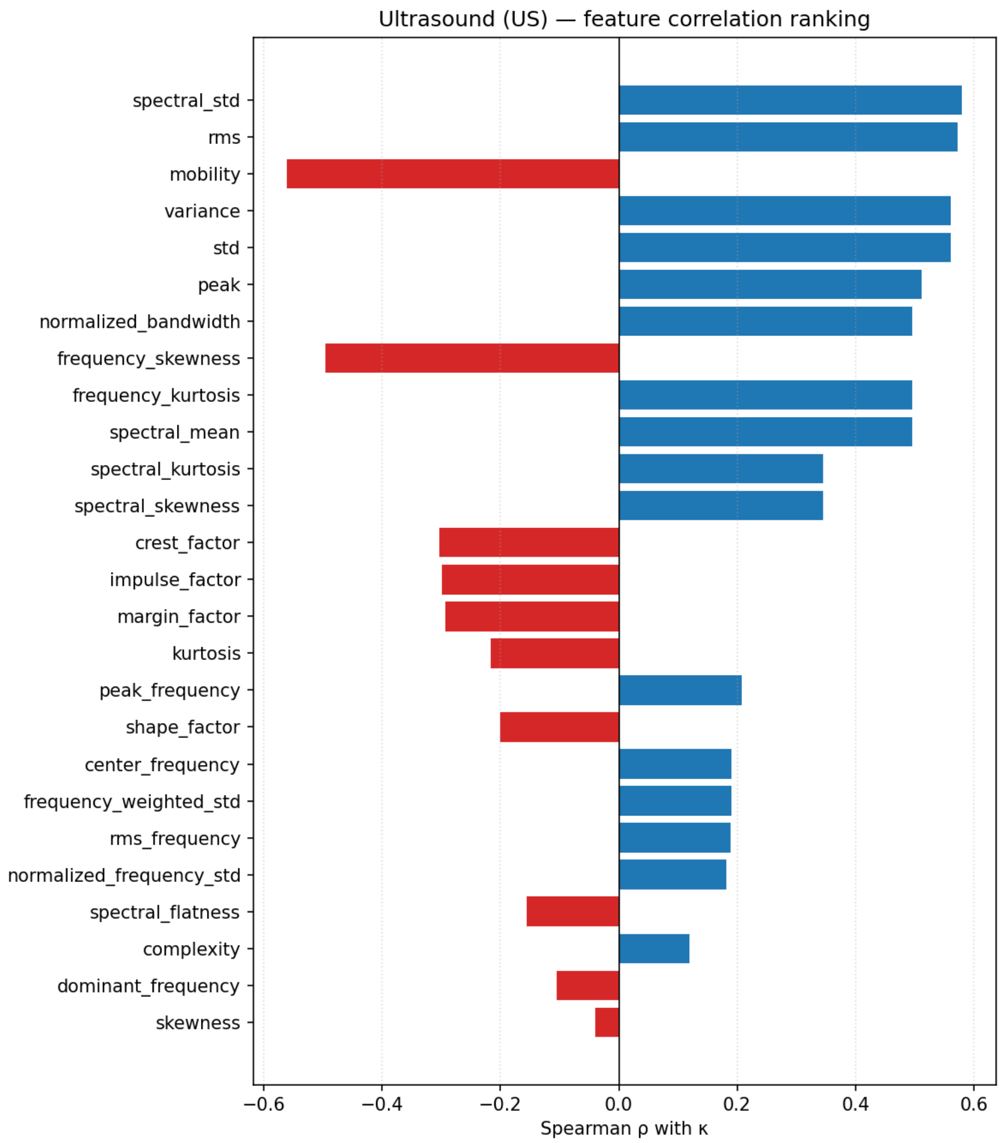

Figure 3: Spearman $|\rho|$ between each acoustic-channel feature and κ, sorted by magnitude. Retained features highlighted.

5.2 Operating-condition confounding analysis

The raw Spearman correlation between broadband signal energy (as a proxy for all amplitude features) and κ is dominated by the shared dependence on rotational speed and temperature. Quantitatively:

- RPM vs. κ: Spearman $\rho = +0.62$, mutual information $= 3.37$ nats. Higher RPM reduces the required viscosity ν_1 (Eq. (2)) and therefore increases κ, while also generating more asperity contact energy and thus higher AE amplitude.

- Temperature vs. κ: Spearman $\rho = -0.68$, mutual information $= 2.65$ nats. Higher temperature reduces actual viscosity ν_0 and therefore reduces κ.

The broadband power of both channels is strongly correlated with RPM: AE band power (0–800 kHz) has $\rho_{\text{RPM}} = +0.90$, while the partial Spearman correlation with κ after controlling for RPM and temperature drops to $\rho_{\text{partial}} = +0.15$. This confirms that broadband signal energy, by itself, conveys very little lubrication-specific information beyond what is already contained in the operating conditions.

5.3 Partial Spearman correlation analysis

To identify features that carry information about κ *beyond* what is explained by RPM and temperature, partial Spearman rank correlations are computed for all 26 features per sensor. The partial correlation $\rho_{\text{partial}}(\text{feature}, \kappa \mid n, T)$ is estimated by regressing out the rank-transformed RPM and temperature from both the feature ranks and the κ ranks, then computing the Spearman correlation of the residuals.

Table 4: Top features by partial Spearman correlation with κ after controlling for rotational speed and temperature. ρ_{raw} is the absolute raw Spearman correlation; ρ_{partial} is the signed partial correlation (positive: higher feature $\rightarrow$ higher κ at constant operating condition). The contrast between raw and partial rankings reveals the extent of operating-condition confounding.

AE			**Acoustic channel**		
Feature	$\lvert\rho_{\text{raw}}\rvert$	ρ_{partial}	Feature	$\lvert\rho_{\text{raw}}\rvert$	ρ_{partial}
Freq. kurtosis	0.450	+0.232	Complexity	0.119	+0.391
Norm. bandwidth	0.450	+0.232	Spectral skewness	0.345	+0.298
Spectral mean	0.450	+0.232	Spectral kurtosis	0.345	+0.298
Variance	0.561	+0.168	Spectral mean	0.496	+0.254
Std	0.561	+0.168	Freq. kurtosis	0.496	+0.254
RMS	0.557	+0.164	Norm. bandwidth	0.496	+0.254
Spectral skewness	0.252	+0.142	Peak	0.511	+0.229
Spectral kurtosis	0.252	+0.142	Std	0.560	+0.221
Spectral std	0.552	+0.129	Variance	0.560	+0.221
Complexity	0.492	+0.014	RMS	0.572	+0.218
Hjorth mobility	0.691	−0.138	Hjorth mobility	0.561	+0.062
Kurtosis	0.109	−0.190			

Key findings from Table 4:

1. *AE frequency distribution features are the most robust κ indicators.* Frequency kurtosis, normalised bandwidth, and spectral mean share the highest AE partial correlations ($\rho_{\text{partial}} \approx +0.23$), suggesting that the distribution of signal energy across the AE frequency spectrum responds to lubrication regime independently of overall signal amplitude.

2. *Amplitude features retain modest independent information.* Variance, std, and RMS retain partial correlations of $\approx +0.16$–0.17, meaning that even after RPM and tem-

perature are controlled, higher signal energy is weakly associated with lower film adequacy.

3. *Acoustic-channel complexity is the strongest independent κ indicator.* Hjorth complexity, ranked only ninth in the raw AE analysis and twelfth in the raw acoustic analysis, rises to the top of the partial ranking for the acoustic channel ($\rho_{\text{partial}} = +0.39$), suggesting that the complexity of the acoustic waveform responds more directly to lubrication conditions than its energy level.

4. *AE Hjorth mobility shows sign reversal after partial control.* While mobility has the highest raw AE correlation ($|\rho| = 0.69$), its partial correlation is negative ($\rho_{\text{partial}} = -0.14$), indicating that the raw correlation is largely an artefact of the shared RPM dependence. At fixed operating conditions, the direction of the mobility–κ relationship reverses, consistent with a complex, non-monotonic interaction between film formation and contact impulsiveness.

5. *AE and acoustic channels respond differently after operating-condition control.* For AE, frequency distribution features dominate the partial ranking; for the acoustic channel, complexity and spectral shape features dominate. This divergence supports the physical distinction between the two sensor modalities (broadband stress waves vs. low-frequency bulk acoustic response).

Figure 4 illustrates the contrast between raw and partial rankings for broadband band power across both sensors.

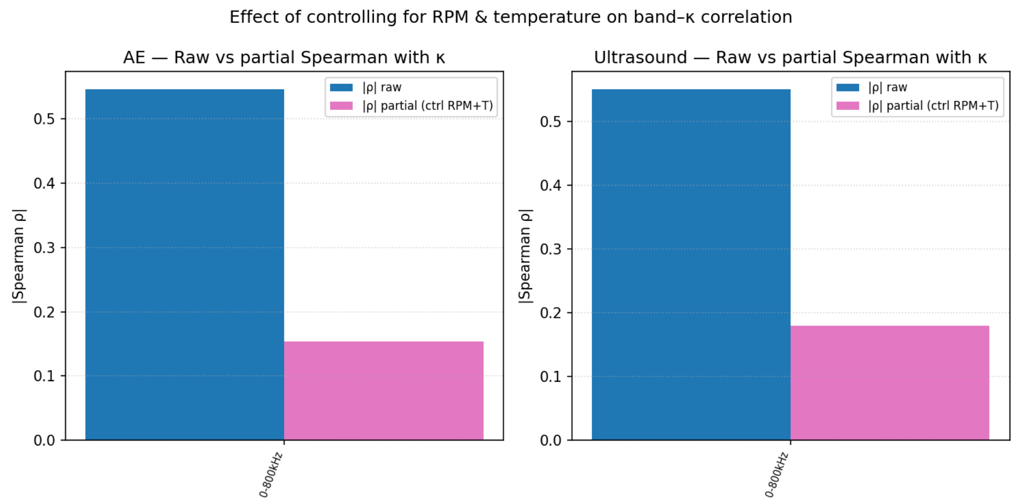

Figure 4: Raw Spearman correlations with RPM (left) and partial Spearman correlation with κ after RPM and temperature control (right) for broadband band power in both sensors. The dramatic reduction from raw to partial underscores the extent of amplitude confounding.

5.4 PCA and regime separability

Figure 5 shows the first two principal components of the retained AE feature set, coloured by κ regime. Boundary and full-film sweeps are broadly separated in the PC1–PC2 plane, while mixed regime sweeps overlap with both. This confirms that the retained feature set carries regime-relevant discriminative information but that the mixed lubrication regime remains the most challenging region, consistent with the known partial physical overlap between boundary and mixed conditions.

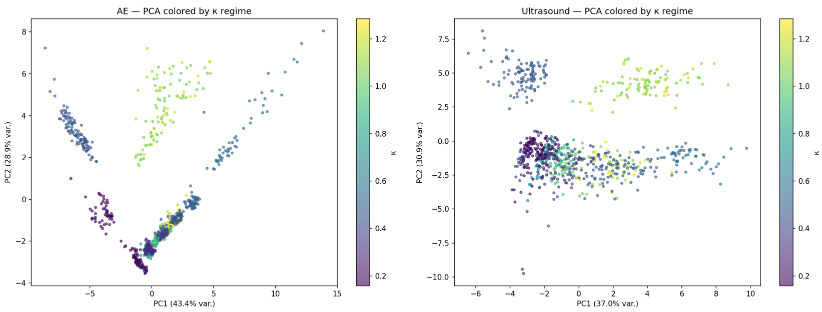

Figure 5: PCA projection of the retained AE feature set (10 features), coloured by lubrication regime. Boundary ($\kappa < 0.5$): red; mixed ($0.5 \leq \kappa < 1$): orange; full film ($\kappa \geq 1$): green.

6 Regression Modelling

6.1 Experimental design

The dataset is divided into a training set (80%, $n_{\text{train}} = 1278$ sweeps across both sensors) and a hold-out test set (20%, $n_{\text{test}} = 320$ sweeps) by random stratified sampling. The hold-out set is withheld entirely from all model development steps and used only for final evaluation.

Four regression model families are evaluated for three sensor configurations (AE only, acoustic channel only, and combined AE+acoustic features):

1. **Elastic Net** (regularised linear regression with combined ℓ_1 and ℓ_2 penalty): hyperparameters α and ℓ_1 ratio tuned by `ElasticNetCV` with 5-fold cross-validation on the training set.

2. **Bayesian Ridge Regression**: EM-based hyperparameter selection; provides predictive uncertainty estimates as a by-product.

3. **Degree-2 Polynomial Regression** (polynomial feature expansion followed by Ridge regression): captures pairwise interactions between features.

4. **LightGBM** (gradient-boosted decision trees): hyperparameters (learning rate, number of leaves, tree depth) tuned by 5-fold cross-validation grid search on the training set.

Model performance is reported as the mean cross-validation R^2 on the training set (5-fold) and the hold-out R^2, MAE, and RMSE on the test set. Confidence intervals on CV R^2 are estimated from the fold-to-fold variance.

6.2 Results

Table 5: Regression model performance across all model families and sensor configurations. CV: mean 5-fold cross-validation R^2 on training set (mean ± fold std). HO: hold-out test set. Best hold-out R^2 in bold.

Model	**Sensors**	**CV R^2**	**HO R^2**	**HO MAE**	**HO RMSE**
LightGBM	AE	0.646	**0.644**	0.136	0.193
LightGBM	Combined	0.632	0.598	0.148	0.205
LightGBM	Acoustic	0.567	0.560	0.143	0.212
Polynomial	AE	0.541	0.537	0.179	0.220
Polynomial	Acoustic	0.446	0.542	0.169	0.217
Polynomial	Combined	0.426	0.465	0.196	0.237
Elastic Net	AE	0.536	0.532	0.185	0.221
Elastic Net	Combined	0.539	0.525	0.187	0.223
Elastic Net	Acoustic	0.418	0.453	0.192	0.237
Bayesian Ridge	AE	0.536	0.532	0.185	0.222
Bayesian Ridge	Combined	0.529	0.519	0.188	0.225
Bayesian Ridge	Acoustic	0.414	0.457	0.190	0.236

Overall performance. The best hold-out performance is achieved by LightGBM on AE features alone ($R^2 = 0.644$, MAE $= 0.136$, RMSE $= 0.193$), corresponding to a hold-out RMSE of approximately 0.19 κ-units on a target range of [0.16, 1.55]. Linear and polynomial models cluster around $R^2 \approx 0.53$–0.54 for AE, consistent with the non-linear feature–κ relationships visible in the PCA projection (Figure 5) and in the scatter plots.

Sensor comparison. AE features consistently outperform acoustic-channel features across all model families (AE R^2: 0.53–0.64 vs. acoustic R^2: 0.42–0.56), reflecting the richer frequency content and more direct sensitivity of AE to asperity contact events.

Sensor fusion does not improve performance. The combined AE+acoustic model achieves $R^2 = 0.598$ at hold-out, which is lower than the AE-only model (0.644) and shows greater degradation from CV to hold-out ($\Delta R^2 = -0.034$) than either single-sensor model, indicating mild overfitting from the larger 21-dimensional feature space. Even Elastic Net, which explicitly regularises the combined feature space, shows no advantage over AE alone (0.525 combined vs. 0.532 AE only).

Model consistency. The close agreement between CV and hold-out R^2 for LightGBM-AE (0.646 vs. 0.644) and Elastic Net-AE (0.536 vs. 0.532) indicates that the model evaluations are stable and that no significant overfitting to the training fold occurred for the single-sensor models.

Figure 6 shows predicted vs. actual κ for the hold-out set for all four model families under the AE-only configuration. Figure 7 shows the fold-to-fold R^2 distributions for the cross-validation stage.

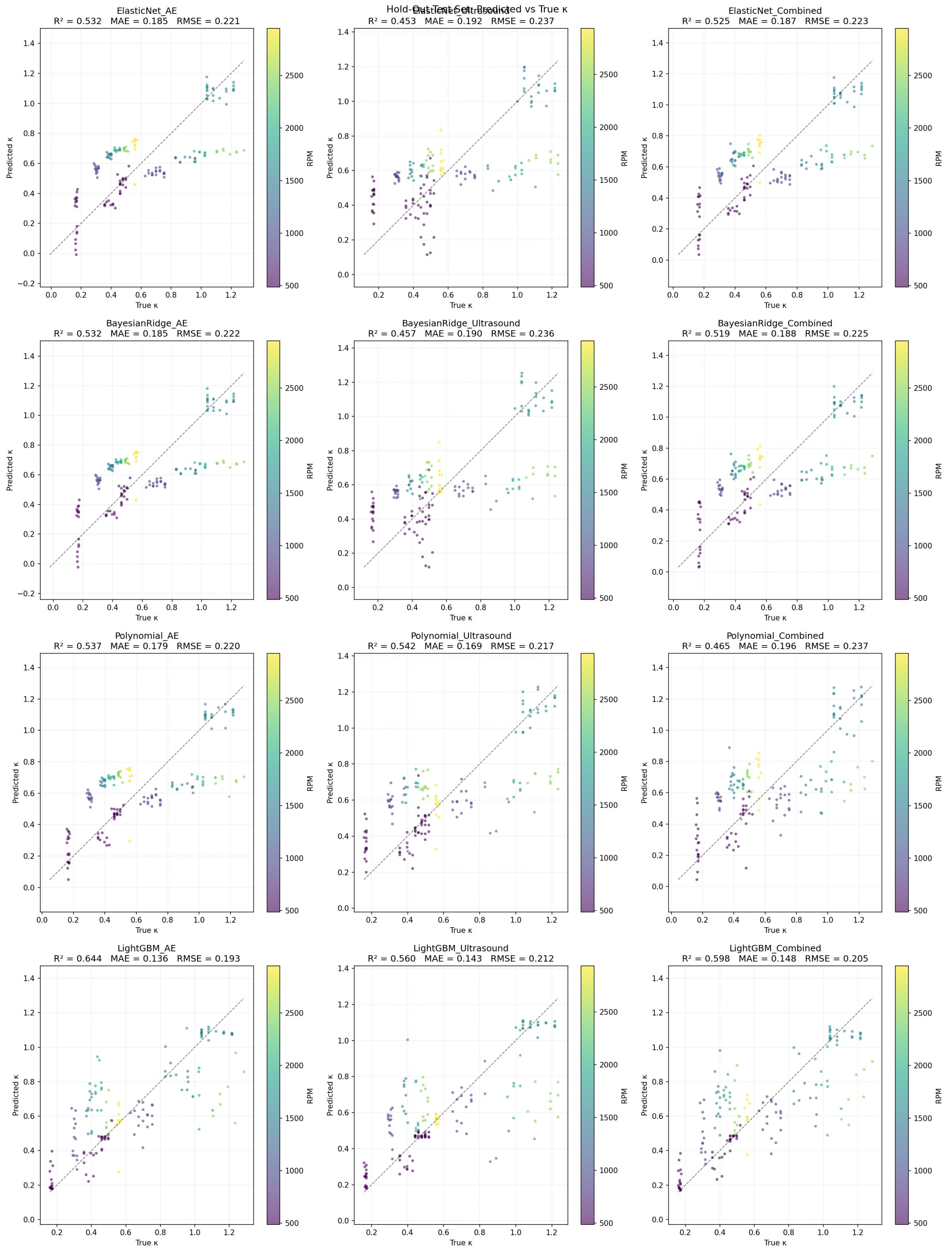

Figure 6: Predicted vs. actual κ on the hold-out test set for all model families and sensor configurations. The dashed diagonal indicates perfect prediction. Residual spread increases in the mixed lubrication regime ($0.5 \leq \kappa < 1.0$).

Figure 7: Cross-validation R^2 distributions across 5 folds for all model–sensor combinations. Fold-to-fold variance is comparable across model families, indicating stable training performance.

7 Discussion

7.1 Interpreting the partial Spearman results

The most striking result of the feature analysis is the *sign reversal* of Hjorth mobility for AE: the highest raw correlation ($|\rho| = 0.69$) becomes a negative partial correlation ($\rho_{\text{partial}} = -0.14$) after operating conditions are controlled. This can be understood as follows. In the raw data, higher RPM is associated with both higher κ (reduced required viscosity per Eq. (2)) and higher signal energy (more rotational contact events per second), so mobility appears to correlate positively with κ. Once RPM is held constant in the partial analysis, the residual variation in κ arises primarily from temperature fluctuations and measurement noise; at a given operating point, the mobility–κ relationship may be weakly negative or noise-dominated. This finding is a cautionary result: a feature that ranks first in a naive Spearman analysis can be substantially or entirely an operating-condition proxy rather than a lubrication condition indicator.

In contrast, frequency distribution features (frequency kurtosis, normalised bandwidth, spectral mean for AE; complexity, spectral skewness/kurtosis for the acoustic channel) show modest but consistent positive partial correlations. These features describe the *shape* of the frequency spectrum rather than its total energy, and are therefore less directly scaled by rotational speed. A physical interpretation is that worsening lubrication shifts signal energy toward higher frequencies or produces a more spectrally non-uniform (higher kurtosis) AE spectrum, reflecting the impulsive, broadband character of asperity contact stress waves. The acoustic-channel complexity result ($\rho_{\text{partial}} = +0.39$) suggests that under boundary lubrication, the bulk acoustic waveform becomes structurally more complex — consistent with multiple overlapping contact transients at different frequencies.

7.2 Why AE outperforms the acoustic channel

The consistently superior performance of AE over the acoustic channel across all model families reflects a fundamental hardware difference. The AE sensor captures genuine wideband stress-wave emissions up to 800 kHz, directly sensitive to individual asperity contact events. The acoustic channel is heterodyned, limiting its useful content to below approximately 10 kHz, where the signal is a mixture of low-frequency machinery noise, VFD harmonics, and structural vibration. The physical information about microscale contact events is largely absent in this frequency range. Replacing the heterodyned probe with a wideband passive acoustic or ultrasound sensor would be expected to substantially improve the acoustic-channel models.

7.3 Why fusion does not help

The negative fusion result ($R^2 = 0.60$ combined vs. 0.64 AE alone) can be explained by two complementary factors. First, the inter-feature correlation analysis shows substantial overlap between the retained AE and acoustic feature sets: both sensors retain mobility, complexity, crest factor, kurtosis, and skewness, and these features carry correlated information. With only 10–11 features per sensor, the combined 21-feature space does not offer sufficient independent information to justify the increased model complexity. Second, the acoustic channel's limited frequency range means its unique contributions are primarily in the low-frequency domain where VFD noise contaminates the signal. If the acoustic channel is effectively a noisy version of low-frequency information already present in AE (given that AE sensors also detect frequencies well below their resonance), adding it to the model introduces noise without adding signal.

This result is an informative contribution to the LCM literature, where AE–ultrasound complementarity is widely *argued* but rarely *tested*. The current finding suggests that complementarity requires both sensors to cover genuinely distinct frequency ranges; with the current hardware, this condition is not met.

7.4 Implications for the κ regression framework

The κ regression framework is validated as a methodologically sound approach for passive non-invasive LCM. A hold-out $R^2 = 0.64$ for a single AE sensor on a difficult, confounded dataset with only two temperature set-points represents a meaningful signal. Comparing this to a naïve baseline (predicting the training mean, $R^2 = 0$) and to the prior work of Jakobsen et al. [1] (who achieved comparable performance with vibration and passive ultrasound), the result is consistent with the expected capability of passive sensing regressed to κ on a small dataset.

The partial Spearman analysis demonstrates that the κ regression framework does not fully remove operating-condition effects from the *features*: amplitude features are still largely RPM proxies even when κ is the regression target. Models trained on κ implicitly learn to use amplitude features as an indirect operating-condition indicator, which inflates apparent performance. Future feature engineering should prioritise spectral shape and ratio features (frequency kurtosis, normalised bandwidth, spectral flatness) that retain information about lubrication state after operating-condition control.

7.5 Limitations

1. **Single bearing and lubricant.** All results are specific to the SYJ25 bearing and Keratech 22 oil. Generalisability to other bearing types, sizes, or lubricants is unknown and is a priority for follow-on experiments.

2. **Heterodyned acoustic channel.** The passive acoustic channel is frequency-limited to $\approx$10 kHz. It should not be equated with wideband ultrasound and results should be interpreted in light of this hardware constraint.

3. **Data gap in the mixed lubrication regime.** With only discrete temperature set-points, the mixed regime ($\kappa \approx 0.5$–0.7) is underrepresented (24% of sweeps). This is precisely the regime of greatest practical interest for early lubrication deterioration warning.

4. **VFD harmonic contamination.** Switching artefacts at $\approx$5 kHz affect low-frequency spectral features of both channels. The partial Spearman analysis reduces but does not eliminate this concern.

5. **Model-derived κ reference.** No independent film thickness measurement (e.g., electrical capacitance or active ultrasonic reflectometry) is available. The κ values are model-derived and subject to the Hamrock–Dowson and Walther equation assumptions.

6. **Dataset size.** With 799 sweeps per sensor from a single bearing, the statistical power to characterise model uncertainty in the mixed regime is limited. The 5-fold CV confidence intervals on R^2 are relatively wide.

8 Conclusion

This paper has presented an exploratory study of simultaneous acoustic emission and passive acoustic sensing regressed to the ISO 281 viscosity ratio κ for non-invasive lubrication condition monitoring of a ball bearing. The main findings are as follows:

1. **Raw Spearman analysis favours amplitude features, but these are predominantly operating-condition proxies.** Hjorth mobility and RMS rank highest in the raw analysis for both sensors, but their partial correlations with κ after controlling for rotational speed and temperature are substantially reduced, with AE mobility exhibiting a sign reversal.

2. **Spectral shape features are the most physically robust κ indicators.** Frequency kurtosis, normalised bandwidth, and spectral mean (AE) and Hjorth complexity and spectral skewness (acoustic channel) show the highest partial Spearman correlations with κ after operating-condition control. Future feature engineering for LCM should prioritise these features over amplitude statistics.

3. **LightGBM on AE features alone achieves the best κ prediction** ($R^2 = 0.644$, RMSE $= 0.193$ on the hold-out set), outperforming linear models ($R^2 \approx 0.53$) and confirming the non-linear character of the feature–κ mapping.

4. **Sensor fusion does not improve upon AE alone** at this dataset scale. The combined model achieves $R^2 = 0.598$, lower than AE alone, attributed to the limited and largely overlapping information content of the heterodyned acoustic channel.

5. **The heterodyned acoustic channel performs below wideband AE** for κ regression, but contributes independent information (particularly via signal complexity) when operating-condition effects are removed. A wideband passive acoustic probe is expected to improve fusion performance substantially.

Collectively, these results demonstrate that passive, non-invasive AE sensing can provide meaningful κ prediction from a single sensor, establish the spectral shape features as the preferred feature class for operating-condition-robust LCM, and provide the first quantitative test of AE–acoustic complementarity for κ regression. The open dataset contributes the first simultaneous AE + passive acoustic benchmark with analytical κ ground truth, directly addressing the absence of standardised LCM datasets identified in the literature.

Future work should address the identified limitations by: acquiring wideband passive ultrasound to replace the heterodyned probe; expanding the dataset to additional lubricant grades, load levels, and bearing types; filling the mixed-regime gap with additional intermediate temperature–speed combinations; and evaluating physically motivated normalised features (e.g., spectral shape ratios, mobility per unit RPM) that are explicitly designed to be operating-condition-invariant.

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Data availability

The complete dataset (HDF5 signal files, Parquet feature tables, and Python analysis pipeline) will be deposited on Zenodo upon acceptance at DOI `[to be assigned]`.

Conflict of interest

The author declares no conflict of interest.

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